Chapter 31 — The Outbox Pattern for Reliable Writes

> **Part IV — Offline-First & Synchronization** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 29 (offline-first — the write path), Ch 15 (idempotency — safe replay), Ch 30 (WorkManager — the drainer), Ch 3 (Room — durable store). Used by every offline write and by Ch 33 (conflicts).
>
> **Reference surface in StreamCart:** reliably delivering cart edits, wishlist changes, posted reviews, and watch-progress to the server — never losing a write even across process death and long offline periods.

31.1 Problem Definition and Real-World Context

In offline-first apps (Ch 29), a local write must be **durably recorded** so it can be sent to the server **later** — after reconnect, after a retry, even after the app is killed. The **outbox pattern** does this: every mutation is written to a durable **outbox** table (in the same local transaction as the optimistic UI update) and a background drainer (Ch 30) replays outbox entries to the server until acknowledged. Without it, offline writes are lost or duplicated:

- **Lost writes.** A write held only in memory (or fired-and-forgotten to the network) vanishes if the request fails and the app is killed — the user's "sent" message / "added to cart" silently disappears.
- **Duplicates.** Retrying a write without a durable record + idempotency key (Ch 15) can create two orders / two comments.
- **Ordering violations.** Two offline edits to the same entity must be applied in order; an ad-hoc approach applies them out of order or drops one.
- **Inconsistent UI vs server.** The optimistic UI update and the "intent to sync" must be atomic — if the UI shows the change but the sync intent wasn't recorded, they diverge.

The outbox makes offline writes **reliable (never lost), exactly-once in effect (via idempotency), and ordered**. It's the write-side backbone of offline-first and a classic distributed-systems pattern (transactional outbox) adapted to the client.

Three practical use cases

1. **E-commerce (StreamCart).** Cart/wishlist edits made offline are queued in the outbox and delivered on reconnect — nothing lost, no duplicate line items.
2. **Social (Social).** A posted comment appears instantly (optimistic) and is queued; it publishes when online, exactly once.
3. **OTT (Watch).** Watch-progress updates accumulate in the outbox (compacted to the latest per title) and sync efficiently.

Why this is architectural

The outbox defines the **durability and ordering guarantees for writes**, the **atomicity of optimistic-update + sync-intent**, and the **replay/idempotency model** — the contract that makes "your offline action is safe" true.

31.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Record every local mutation durably (outbox) atomically with the optimistic update.
F2	Each outbox entry carries an idempotency key (Ch 15).
F3	Drain the outbox to the server in order (per entity), removing entries on ack.
F4	Retry transient failures; stop on permanent failures (dead-letter).
F5	Survive process death / reboot (durable).
F6	Compact redundant operations (optional) to reduce sync volume.

Non-functional requirements

Attribute	Policy
Reliability	No lost writes; durable; survives kill/reboot.
Correctness	Exactly-once effect (idempotency); ordered per entity.
Performance	Batched drain; compaction; bounded size.
Observability	Entry status/age; dead-letter visibility.
Consistency	Optimistic update + outbox entry atomic (one transaction).

Scope

- **In scope:** the outbox table + entry model, atomic enqueue with the optimistic update, ordered durable drain, idempotency, retry/dead-letter, compaction, status.
- **Out of scope:** the sync *orchestration* (Ch 29/30), delta *pull* (Ch 32), conflict resolution (Ch 33), idempotency-key semantics (Ch 15).

Assumptions & constraints

- Room (Ch 3) for durability; WorkManager (Ch 30) as the drainer; idempotency keys (Ch 15); backend applies mutations idempotently. `minSdk 24`.

31.3 System Design Fundamentals

- **Transactional outbox.** The mutation's **optimistic local effect** (updating the entity's row) and the **outbox entry** (the intent to sync) are written in **one database transaction** — so they're consistent: the UI never shows a change whose sync intent was lost, and vice versa. This atomicity is the pattern's core.
- **Durable queue.** The outbox is a Room table (`pending_mutations`), so entries survive process death and reboot — the drainer picks up where it left off.
- **Ordered drain (per entity).** Entries are processed in insertion order (FIFO), at least per entity, so sequential edits apply correctly. A global strict order is possible but per-entity ordering is usually sufficient and more parallelizable.
- **Idempotency (Ch 15).** Each entry has an idempotency key; the server dedups, so a retried/replayed entry applies exactly once. Delivery is at-least-once; effect is exactly-once.
- **Retry & dead-letter.** Transient failures leave the entry for retry (with backoff via WorkManager, Ch 30); a permanent failure (validation/4xx) moves the entry to a **dead-letter** state (or marks the entity FAILED) and surfaces to the user — never silently dropped, never retried forever.
- **Compaction (optional).** Redundant ops can be merged before sending: multiple updates to the same field → keep the latest; a create followed by a delete → cancel both locally. Reduces sync volume for chatty updates (e.g., watch progress).
- **Entry lifecycle.** `PENDING → SENDING → (acked: removed) / (retryable: back to PENDING) / (permanent: DEAD_LETTER)` .

Core insight: **write the optimistic update and a durable, idempotent outbox entry in one transaction, then drain the outbox in order with retry/dead-letter.** This guarantees no lost writes, exactly-once effects, and correct ordering — the reliability contract of offline writes.

31.4 High-Level Architecture



Components & ownership

- **Outbox table** (Room) durably stores entries; entity update + entry are **atomic**.
- **Drainer** (`SyncWorker` , Ch 30) reads PENDING entries in order, sends them idempotently, and acks/retries/dead-letters.
- **Compaction** (optional) merges redundant entries before send.
- **Dead-letter** surfaces permanent failures (Ch 29).

Direction: write → (entity + outbox entry, one tx) → drainer → server → ack/retry/dead-letter.

31.5 Detailed Component Design



Element	Responsibility	Note
`OutboxEntry`	one pending mutation	seq (order), idempotency key, type, entityId, payload
atomic enqueue	entity update + entry in one tx	consistency (F1)
`OutboxDao`	durable queue access	ordered reads, ack, status
`Drainer`	send in order, ack/retry/dead-letter	idempotent (Ch 15), WorkManager (Ch 30)
compaction	merge redundant	reduce volume

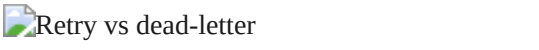
`seq` **(monotonic insertion order)** gives a total order for the drain; per-entity ordering is derived by filtering on `entityId`.

31.6 End-to-End Data Flow

Atomic optimistic write + outbox, then drain


Atomic optimistic write + outbox, then drain

Retry vs dead-letter


Retry vs dead-letter

31.7 Complete Kotlin Implementation

Package: `com.streamcart.sync.outbox`.

31.7.1 Outbox entry + DAO

```
enum class OutboxStatus { PENDING, SENDING, DEAD_LETTER }

@Entity(tableName = "outbox")
data class OutboxEntry(
    @PrimaryKey(autoGenerate = true) val seq: Long = 0,     // insertion order
    val idempotencyKey: String,                             // Ch 15
    val type: String,                                       // "cart.add"
    val entityId: String,
    val payload: String,                                    // serialized mutation
    val status: OutboxStatus = OutboxStatus.PENDING,
    val attempts: Int = 0,
    val createdAt: Long,
)

@Dao
interface OutboxDao {
    @Insert suspend fun insert(entry: OutboxEntry)
    @Query("SELECT * FROM outbox WHERE status = 'PENDING' ORDER BY seq ASC LIMIT :limit")
    suspend fun pending(limit: Int = 50): List<OutboxEntry>
    @Query("DELETE FROM outbox WHERE idempotencyKey = :key") suspend fun remove(key: String)
    @Query("UPDATE outbox SET status = :s, attempts = attempts + 1 WHERE idempotencyKey = :key")
    suspend fun bump(key: String, s: OutboxStatus)
    @Query("SELECT COUNT(*) FROM outbox WHERE status = 'PENDING'") fun pendingCount(): Flow<Int>
    // Compaction example: keep only the latest progress update per entity.
    @Query("DELETE FROM outbox WHERE type = 'progress.update' AND entityId = :id AND seq <
:keepSeq")
    suspend fun compactProgress(id: String, keepSeq: Long)
}
```

31.7.2 Atomic enqueue (optimistic update + outbox entry in one transaction)

```
class CartRepositoryImpl @Inject constructor(
    private val db: StreamCartDatabase,
    private val cartDao: CartDao,
    private val outbox: OutboxDao,
    private val syncScheduler: SyncScheduler,       // Ch 30
    private val json: Json, private val clock: Clock,
) : CartRepository {

    override suspend fun addItem(productId: String) {
        val item = CartItemEntity(id = UUID.randomUUID().toString(), productId = productId,
            quantity = 1, updatedAt = clock.now(), syncStatus = SyncStatus.PENDING)
        db.withTransaction {                                  // ATOMIC: both or neither (F1)
            cartDao.upsert(item)                              // optimistic UI update
            outbox.insert(OutboxEntry(
                idempotencyKey = UUID.randomUUID().toString(),   // Ch 15
                type = "cart.add", entityId = item.id,
                payload = json.encodeToString(item.toRequest()), createdAt = clock.now(),
            ))
        }
        syncScheduler.requestSync()                           // nudge drainer (Ch 30)
    }
}
```

31.7.3 Drainer (ordered, idempotent, retry/dead-letter)

```
class OutboxDrainer @Inject constructor(
    private val outbox: OutboxDao,
    private val api: MutationApi,
    private val entityUpdater: EntityUpdater,       // marks entity SYNCED/FAILED
) {
    suspend fun drain(): DrainResult {
        var anyRetryable = false
        for (entry in outbox.pending()) {                     // ordered by seq
            outbox.bump(entry.idempotencyKey, OutboxStatus.SENDING)
            when (val r = api.apply(entry.type, entry.idempotencyKey, entry.payload)) {  //
idempotent (Ch 15)
                is Outcome.Success -> { outbox.remove(entry.idempotencyKey);
entityUpdater.markSynced(entry.entityId, r.data) }
                is Outcome.Failure -> when {
                    r.error.isRetryable() && entry.attempts < MAX_ATTEMPTS -> {
                        outbox.bump(entry.idempotencyKey, OutboxStatus.PENDING); anyRetryable =
true
                    }
                    else -> {   // permanent OR exhausted -> dead-letter, surface
                        outbox.bump(entry.idempotencyKey, OutboxStatus.DEAD_LETTER)
                        entityUpdater.markFailed(entry.entityId, r.error)
                    }
                }
            }
        }
        return if (anyRetryable) DrainResult.Retry else DrainResult.Done   // -> WorkManager
Result (Ch 30)
    }
    private companion object { const val MAX_ATTEMPTS = 8 }
}
```

31.7.4 Compaction (optional, before drain)

```
// For chatty updates (watch progress), keep only the latest per entity.
suspend fun compact(outbox: OutboxDao, entityId: String) {
    val latest = outbox.pending().filter { it.type == "progress.update" && it.entityId ==
entityId }
        .maxByOrNull { it.seq } ?: return
    outbox.compactProgress(entityId, keepSeq = latest.seq)   // delete older redundant entries
}
```

31.7.5 Pending count for UI (Ch 29 sync status)

```
val pendingCount: Flow<Int> = outbox.pendingCount()          // "2 changes will sync"
```

31.7.6 Module structure

```
:sync/outbox   OutboxEntry, OutboxDao, OutboxDrainer, compaction
:core:database outbox table (Room, Ch 3)
:sync          SyncWorker calls drainer (Ch 30); SyncScheduler
```

31.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Optimistic update without outbox entry (or vice versa)	UI/server divergence	**one transaction** (`withTransaction`) — atomic (F1)
Two drains run concurrently	double send	`enqueueUniqueWork` (Ch 30); `SENDING` status guard
Write enqueued during drain	missed/ordered wrong	`seq` ordering; new entry drained next pass
Retry re-sends	duplicate effect	idempotency key (Ch 15) — server dedups
Process death mid-drain	partial	durable entries; `SENDING` reset to `PENDING` on next run (or idempotent re-send)
Ordering across entities	inter-entity dependency	per-entity FIFO (seq); model dependencies explicitly if needed

- **Atomic enqueue is the core invariant:** the optimistic update and the outbox entry must commit together. A `SENDING` entry interrupted by process death is safely re-sent next run because the server dedups by key (Ch 15).
- **Unique drain work** (Ch 30) prevents two drainers double-sending.

31.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Offline	send fails (network)	leave PENDING; retry on reconnect	no loss
Transient server error (5xx/timeout)	retryable	attempts++, backoff (Ch 30)	eventual success
Permanent error (4xx/validation)	non-retryable	DEAD_LETTER + mark entity FAILED; surface (Ch 29)	user notified, not dropped
Process death mid-send	durable outbox	re-send next run (idempotent)	one effect
Attempts exhausted	attempts > max	dead-letter; surface	user can retry/discard
Chatty updates flood outbox	growth	compaction (keep latest)	bounded
Entry stuck SENDING	crash after mark	reset SENDING→PENDING on startup	re-drained
Dependency between mutations	out-of-order apply	per-entity order; model deps	correct



> **Never silently drop a write.** A permanent failure goes to a **dead-letter** state the user can see and act on (retry/discard) — the outbox's promise is that the user's action is either delivered or visibly flagged, never lost.

31.10 Scaling the Implementation

- **Level 1.** A single outbox table + atomic enqueue + a simple drainer.
- **Level 2.** Ordered drain + idempotency (Ch 15) + retry/backoff via WorkManager (Ch 30).
- **Level 3 — production.** Dead-letter + user surfacing; pending-count UX; per-entity ordering; `SENDING` recovery; compaction for chatty updates.
- **Level 4 — large.** Batched sends (one request, many entries); priority entries (payments before analytics, Ch 14); per-domain outboxes; backpressure on huge outboxes; encrypted outbox (Ch 24).
- **Level 5 — enterprise.** Transactional outbox mirroring server patterns, exactly- once pipelines, conflict-aware draining (Ch 33), outbox observability/SLAs, cross-device de-dup.

Avoid overengineering: apps that only need offline *reads* (Ch 3) don't need an outbox. Add it when users make writes that must survive offline/kill — it's the right tool there, overkill otherwise.

31.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Write durability	Outbox (durable table)	in-memory queue	Outbox always for offline-first (survives kill); in-memory only for ephemeral
Atomicity	optimistic + entry in one tx	separate writes	One tx — avoid UI/server divergence
Ordering	per-entity FIFO (seq)	global strict order	Per-entity usually enough + parallel; global for cross-entity deps
Delivery	at-least-once + idempotency	exactly-once transport	At-least-once + keys (achievable); exactly-once transport is impossible
Drain	WorkManager (Ch 30)	foreground loop	WorkManager (survives kill, constrained)
Redundant ops	compaction	send all	Compact chatty updates; send all for distinct ops
Failure	dead-letter + surface	retry forever / drop	Dead-letter — bounded + visible

Outbox vs fire-and-forget / in-memory queue: fire-and-forget loses writes on failure; an in-memory queue loses them on process death. The durable outbox is the only approach that guarantees no lost writes across offline periods and kills — at the cost of a table + drainer + idempotency discipline. For offline-first writes, it's the correct pattern.

Compaction vs send-all: compaction reduces volume for chatty, last-writer-wins updates (progress, typing, cursor) but adds logic and can lose intermediate history; send-all preserves every operation (needed for event-sourced/auditable data). Compact only where only the latest matters.

31.12 Security Considerations

- **Encrypt the outbox** (Ch 24) if entries contain sensitive payloads — it's user data persisted on device, exposed on a stolen/rooted device otherwise.
- **Idempotency keys** (Ch 15) prevent duplicate side effects on replay — critical for financial mutations in the outbox.
- **Server authorizes every drained mutation** (Ch 29) — a tampered client could insert arbitrary outbox entries; the backend re-validates authz per entry.
- **Clear the outbox on logout** (Ch 22) and scope per account (Ch 42) — pending mutations are the user's data/intent.
- **Don't log payloads** (Ch 28) — they may contain PII/secrets.
- **Bound dead-letter exposure:** dead-lettered sensitive mutations shouldn't linger unencrypted; purge per policy.
- Client vs backend: the outbox guarantees *client-side* delivery intent; the backend remains the authority on what's actually applied and authorized.

31.13 Performance and Resource Optimization

- **Batch sends:** drain many entries per sync run / one batched request where the server supports it — fewer round trips (Ch 14/30).
- **Compact** chatty updates to cut volume.
- **Bound the outbox:** cap size; backpressure or coalesce if it grows (long offline).
- **Index** `status` / `seq` for fast ordered reads.
- **Drain under constraints** (Ch 30): on connectivity/charging, batched — not per keystroke.
- **Profiling:** outbox size, drain duration, pending-entry age, dead-letter rate, sync round trips.

31.14 Testing Strategy

In-memory Room + fake API; test atomicity, ordering, idempotency, retry/dead-letter.

```
class OutboxTest {
    @Test fun `optimistic update and outbox entry commit atomically`() = runTest {
        repo.addItem("p1")
        assertEquals(1, cartDao.count())              // entity
        assertEquals(1, outbox.pending().size)        // entry — both present
    }

    @Test fun `drain sends in order and removes on ack`() = runTest {
        repo.addItem("p1"); repo.addItem("p2")
        val api = RecordingApi(success = true)
        OutboxDrainer(outbox, api, updater).drain()
        assertEquals(listOf("cart.add", "cart.add"), api.sentTypes)
        assertTrue(outbox.pending().isEmpty())
    }

    @Test fun `transient failure keeps entry for retry`() = runTest {
        repo.addItem("p1")
        OutboxDrainer(outbox, RecordingApi(ioError = true), updater).drain()
        assertEquals(1, outbox.pending().size)        // retried later
    }

    @Test fun `permanent failure dead-letters and surfaces`() = runTest {
        repo.addItem("p1")
        OutboxDrainer(outbox, RecordingApi(code = 422), updater).drain()
        assertEquals(OutboxStatus.DEAD_LETTER, outbox.statusOf("p1"))
        assertEquals(SyncStatus.FAILED, cartDao.findById("p1")!!.syncStatus)
    }

    @Test fun `double drain does not double-apply (idempotent)`() = runTest {
        repo.addItem("p1")
        val api = CountingApi()
        val d = OutboxDrainer(outbox, api, updater); d.drain(); d.drain()
        assertEquals(1, api.appliedKeys.size)          // server dedups by key (Ch 15)
    }

    @Test fun `compaction keeps only latest progress`() = runTest {
        repeat(5) { repo.updateProgress("t1", it) }
        compact(outbox, "t1")
        assertEquals(1, outbox.pending().count { it.type == "progress.update" })
    }
}
```

Also: process-death resume (entries persist; SENDING→PENDING recovery), per-entity ordering, encrypted outbox (Ch 24), clear on logout.

31.15 Production Observability

- **Metrics:** outbox size, pending-entry age (how long writes wait — SLO), drain success/retry/dead-letter rate, batch size, write-loss incidents (0), compaction savings.
- **Logs:** `outbox{op, type, status, attempts}` (ids/types, no payload PII, Ch 28), correlation IDs (Ch 1/84).
- **Debugging:** writes lost → non-atomic enqueue / non-durable outbox; duplicates → missing idempotency (Ch 15); stuck SENDING → crash recovery not resetting status; outbox growing → drain blocked (constraints/failures) or no compaction; dead-letter pile-up → backend rejecting (investigate).
- **Alerting:** rising pending-entry age or dead-letter rate → sync/backend incident.
- **Privacy:** outbox entries are user data — redact logs; clear on logout/deletion.

31.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *What is the outbox pattern?* Durably record each local mutation and replay it to the server until acked.
2. *Why durable (Room)?* Survive process death/offline so writes aren't lost.
3. *What's atomic?* The optimistic entity update + the outbox entry (one transaction) — no UI/server divergence.
4. *Delivery semantics?* At-least-once delivery + idempotency (Ch 15) → exactly-once effect.
5. *Entry lifecycle?* PENDING → SENDING → removed(ack) / PENDING(retry) / DEAD_LETTER(permanent).

Ordering/idempotency

6. *Ordering guarantee?* Per-entity FIFO via `seq`.
7. *Why idempotency keys?* Retries/replays must apply once (Ch 15).
8. *Two offline edits same entity?* Ordered drain applies in sequence.

Failure handling

9. *Transient failure?* Leave PENDING; retry with backoff (Ch 30).
10. *Permanent failure?* Dead-letter + surface; never drop; never retry forever.
11. *Process death mid-send?* Durable; re-send next run (idempotent); reset SENDING.
12. *Attempts cap?* Dead-letter after max attempts.

Optimization

13. *Compaction?* Merge redundant ops (keep latest progress) to cut volume.
14. *Batching?* Drain many entries per run / one request.
15. *Bounding?* Cap outbox; backpressure on long offline.

Integration

16. *Who drains it?* WorkManager (Ch 30) — survives kill, constrained.

17. *Relation to offline-first?* It's the write path (Ch 29).
18. *Relation to conflicts?* Drained mutations may conflict with remote (Ch 33).

Security

19. *Encrypt the outbox?* Yes if payloads sensitive (Ch 24).
20. *Authorization?* Backend authorizes each drained mutation (Ch 29).
21. *Logout?* Clear outbox; per-account (Ch 22/42).

Testing

22. *How test atomicity?* Assert entity + entry both present after a write.
23. *How test dead-letter?* Permanent error → DEAD_LETTER + entity FAILED.
24. *How test idempotency?* Double drain → one applied key.

Production debugging

25. *Writes lost?* Non-atomic/non-durable outbox.
26. *Duplicates?* Missing idempotency.
27. *Outbox growing?* Drain blocked or no compaction.

Design-change challenges

1. **Chatty progress updates flood sync.** Compaction (keep latest).
2. **Payments in the outbox must not double-charge.** Idempotency keys (Ch 15).
3. **Surface failed writes to the user.** Dead-letter + mark entity FAILED (Ch 29).
4. **Batch outbox sends.** Server batch endpoint; drain in batches.
5. **Encrypt pending sensitive mutations.** Encrypted outbox (Ch 24).

Mock walkthrough (abridged)

Clarify (offline writes? durability? ordering?) → outbox table + atomic enqueue (entity + entry in one tx) → idempotency key (Ch 15) → ordered drain via WorkManager (Ch 30) → retry/dead-letter → compaction/batching → security (encrypt/clear/authz) → testing.

31.17 Practical Exercises

1. **Beginner.** Add an outbox table + atomic enqueue (optimistic update + entry in one transaction). *Eval:* both present; UI instant.
2. **Intermediate.** Build an ordered drainer with idempotency keys; drain via WorkManager on connectivity. *Eval:* in-order; no dup on double drain.
3. **Advanced.** Add retry/dead-letter + surface failures + compaction for chatty updates. *Eval:* transient retried; permanent dead-lettered; volume reduced.
4. **Production debugging.** Fix an app that loses offline writes (non-atomic/in- memory) and duplicates (no idempotency). *Eval:* durable, atomic, idempotent.
5. **System design challenge.** Batch + prioritize the outbox (payments before analytics) with an encrypted store. *Eval:* fewer round trips; priority honored; encrypted.

(Solutions withheld unless requested.)

31.18 Chapter Summary & Reference Sheet

Key decisions: record every local mutation in a **durable outbox** (Room), writing the **optimistic entity update and the outbox entry in one transaction** (atomic); give each entry an **idempotency key** (Ch 15); **drain in order** (per-entity `seq`) via **WorkManager** (Ch 30), removing on ack; **retry transient** failures with backoff and **dead-letter permanent** ones (surface, never drop); **compact** chatty redundant ops; **encrypt + clear** the outbox (Ch 24/22); the **backend authorizes** each drained mutation. **No lost writes, exactly-once effect, correct order.**

Concepts: transactional outbox, durable queue, atomic optimistic-update+entry, at-least-once + idempotency, per-entity ordering, retry/dead-letter, compaction, entry lifecycle.

Common mistakes: in-memory/fire-and-forget writes (lost); non-atomic enqueue (UI/server divergence); no idempotency (duplicates); retrying forever / silently dropping (no dead-letter); unordered drain; unbounded/un-compacted outbox; unencrypted sensitive entries; not clearing on logout.

Checklist:

- [ ] durable outbox (Room)
- [ ] atomic entity+entry (one tx)
- [ ] idempotency key (Ch 15)
- [ ] ordered drain (seq)
- [ ] WorkManager drainer (Ch 30)
- [ ] retry transient + dead-letter permanent
- [ ] surface failures (Ch 29)
- [ ] compaction for chatty ops
- [ ] batch sends
- [ ] encrypt + clear (Ch 24/22)
- [ ] server authorizes.

Key APIs/patterns: Room `@Transaction` / `withTransaction`, outbox table + `OutboxDao`, `OutboxStatus` lifecycle, idempotency keys (Ch 15), WorkManager drainer (Ch 30), compaction queries, `pendingCount` Flow.

References:

- Transactional outbox pattern: https://microservices.io/patterns/data/transactional-outbox.html
- Offline-first (writes): https://developer.android.com/topic/architecture/data-layer/offline-first
- Room transactions (Ch 3): https://developer.android.com/training/data-storage/room
- (composes) Ch 15 Idempotency, Ch 29 Offline-First, Ch 30 WorkManager, Ch 33 Conflicts

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; outbox entry + atomic enqueue + ordered drainer + compaction + pending count; concurrency (atomicity, unique drain, SENDING recovery); failure (retry/dead- letter, no silent loss); 5 levels; outbox-vs-in-memory and compaction-vs-send-all trade-

offs; security (encrypt/clear/authz); perf (batch/compact/bound); atomicity/ idempotency tests; observability (pending age); 27 Q&A + 5 challenges. ✅

- **Expand later:** WorkManager drain (Ch 30), conflicts on drained mutations (Ch 33), batching/priority (Ch 14), encryption (Ch 24).
- **Interview coverage:** exceeds ≥25 bar.